lsqvce

Least-squares adjustment with variance component estimation

Theory, practical considerations, implementation, error handling

This document collects the theory the toolbox rests on and the reasons for every place where the implementation deviates from the textbook. It is written for a reader who knows least squares and wants to know exactly what lsqvce computes, why, and where it refuses to compute. Each statement about the toolbox names the work package (WP) that pinned it, so the claim can be checked against HANDOVER.md rather than believed.

Sources. Koch (1999); Teunissen (2000); Förstner (1979); Koch and Kusche (2002); Kusche (2003); Teunissen and Amiri-Simkooei (2008); Amiri-Simkooei (2016); Ellmer (2018); Kvas and Mayer-Gürr (2019); Kvas et al. (2019); Golub and Van Loan (2013); Higham (2002). Full list in section 10.

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1 The model and the two ways to solve it

The linear Gauss-Markov model relates n reduced observations $\mathbf{l}$ (observed minus computed) to u unknowns $\mathbf{x}$:

$$\mathbf{l} = \mathbf{A}\mathbf{x} + \mathbf{e},\quad E\{\mathbf{e}\} = 0,\ D\{\mathbf{e}\} = \sigma_0^2\mathbf{Q} = \sigma_0^2\mathbf{P}^{-1} \tag{1.1}$$

The best linear unbiased estimator is the weighted least-squares solution and its cofactor matrix

$$\hat{x} = (\mathbf{A}^T\mathbf{P}\mathbf{A})^{-1}\mathbf{A}^T\mathbf{P}\mathbf{l},\quad \mathbf{Q}_{xx} = (\mathbf{A}^T\mathbf{P}\mathbf{A})^{-1} \tag{1.2}$$

with residuals $\mathbf{v} = \mathbf{l} - \mathbf{A}\hat{\mathbf{x}}$ and the a-posteriori variance of unit weight

$$\hat{\sigma}_0^2 = \mathbf{v}^T\mathbf{P}\mathbf{v} / r,\quad r = n - u \tag{1.3}$$

Everything the toolbox reports about quality descends from (1.3): the redundancy r is the budget of degrees of freedom that the variance components will later divide among observation groups (section 3). Sign convention pinned in the toolbox: $\mathbf{l}$ is O − C and $\mathbf{v} = \mathbf{l} - \mathbf{A}\mathbf{x} - \mathbf{B}\mathbf{y}$ (residuals.m; DATA_CONTRACT §3).

1.1 Normal equations versus QR

Two backends solve (1.2). The normal-equation (NEQ) path forms $\mathbf{N} = \mathbf{A}^T\mathbf{P}\mathbf{A}$ and factors it by Cholesky; cost about $nu^2/2 + u^3/6$, memory $O(u^2)$, and -- decisively for satellite gravimetry -- the normal matrices of independent arcs and systems ADD, so accumulation is a reduction and combination is a sum. The price is the condition number:

$$\kappa(\mathbf{A}^T\mathbf{P}\mathbf{A}) = \kappa(\mathbf{P}^{1/2}\mathbf{A})^2 \tag{1.4}$$

Once $\kappa(\mathbf{A})$ exceeds roughly 10^7-10^8, $\mathbf{N}$ is numerically singular in IEEE double precision (Golub and Van Loan §5.3; Higham §20). The orthogonal (QR) path factors $\mathbf{P}^{1/2}\mathbf{A} = \mathbf{Q}\mathbf{R}$ and solves $\mathbf{R}\mathbf{x} = \mathbf{Q}^T\mathbf{P}^{1/2}\mathbf{l}$; multiplication by an orthogonal matrix neither stretches vectors nor amplifies error, so the forward error grows with $\kappa(\mathbf{A})$ plus a residual-weighted κ^2 term instead of κ^2 outright. It costs about twice the flops and does not add across blocks.

> Pinned in lsqvce (WP-7, WP-49/50, R4 §11.2 item 2): BOTH backends exist behind one interface (QRSystem, NormalEqSystem). The engines default to "qr"; the streaming accumulator merges QR factors of arcs and days ("mergeRaw"), so the arc-wise accumulation the NEQ path is famous for is available under QR as well. Changing the default is a decision, not an option (Options.Backend documents it).

1.2 Local unknowns and the block-arrow structure

Satellite data carry unknowns that belong to one arc only -- initial states, accelerometer biases, empirical accelerations. With $\mathbf{B}$ the design of these *local* unknowns $\mathbf{y}$ the normal system of one observing system is block-arrow: a common block $\mathbf{N}_{cc}$, a local block $\mathbf{N}_{ll}$ and the coupling $\mathbf{N}_{lc}$. Pre-elimination by the Schur complement removes the local unknowns exactly:

$$\mathbf{N}_{cc}^{\text{red}} = \mathbf{N}_{cc} - \mathbf{N}_{lc}^T\mathbf{N}_{ll}^{-1}\mathbf{N}_{lc} \tag{1.5}$$

so the global system carries only the common unknowns; the local ones are recovered afterwards by back-substitution. Two consequences matter in practice. First, any regularisation of $\mathbf{N}_{ll}$ biases $\mathbf{x}_c$ through (1.5) -- damping the local block is never free (BlockAccumulator help, "REGULARISATION BIASES THE COMMON BLOCK"). Second, a system with no local unknowns is a valid system: the Schur step is the identity (G-07, decided 2026-08-31).

1.3 Nonlinear models

Nonlinear observation equations are linearised at approximate values $\mathbf{x}_0$ and iterated (Gauss-Newton): the design becomes the Jacobian at the current iterate and $\mathbf{l}$ the reduced observation $\mathbf{l} - f(\mathbf{x})$. The toolbox iterates with optional Levenberg damping and stops on the step or the cost (solveNonlinear; through fit as the batched kind with *x0*, WP-56b).

2 Ill-conditioning, rank deficiency, regularisation

Spherical-harmonic gravity normal matrices are ill-conditioned by nature: signal power decays with degree, polar gaps and uneven sampling make high orders nearly unobservable, and the parameters differ by many orders of magnitude. Three tools address this.

- **Scaling.** Symmetric Jacobi scaling **DND** with **D** = diag(**N**)$^{-1/2}$ equalises the diagonal and lowers κ before any factorisation. Cheap, exact (the solution is rescaled back), always worth doing.
- **Rank-revealing detection.** A Cholesky factorisation that fails, or a pivoted QR / eigenvalue ratio near machine precision, signals a datum defect or a broken stochastic model. The toolbox detects datum defects from the eigenvalue ratios of the reduced system and reports the redundancy accordingly (datum.m).
- **Regularisation.** Tikhonov / Kaula regularisation adds a prior block, (**N** + λ**K**)**x** = **n**. The weight λ can be chosen by the L-curve, by generalised cross-validation (gcvLambda), or -- most consistently -- by treating the prior as one more observation group whose variance component the VCE estimates (Koch and Kusche 2002; Xu et al. 2006). This is WP-24 (CommonPrior, applyRegularisation): the prior enters as observations and section 3 weights it.

> Never add a silent diagonal "nugget" to make a factorisation succeed. A regularisation that was not declared is a result that cannot be interpreted. lsqvce records regUsed / lambdaUsed in every system diagnostic and refuses to hide the choice.

2.1 Cholesky failure is a signal

MATLAB's two-output form [R, p] = chol(N) returns p > 0 instead of throwing. That is the right primitive: the failure is information (rank deficiency, an indefinite covariance, a mis-scaled group), to be reported and, where a fallback exists, handled by an orthogonal factorisation -- not masked. Section 7 returns to this under error handling: what lsqvce does when the boundary itself is fuzzy (a to-eps singular matrix whose Cholesky "succeeds" on one BLAS and fails on another).

3 Variance component estimation

Write the dispersion of the observations as a linear combination of known cofactor matrices with unknown variance components:

$$D\{\mathbf{l}\} = \Sigma_k \, \sigma_k^{\,2} \, \mathbf{Q}_k \tag{3.1}$$

The components $\sigma_k^{\,2}$ are what VCE estimates. Three rigorous families exist; the toolbox implements the first and its Monte-Carlo form.

3.1 Helmert and the redundancy-contribution form

Helmert's (1907) estimator requires the traces of products of cofactor matrices with the inverse normal matrix. Förstner (1979) and Koch reduce it, for uncorrelated groups, to the iterated form

$$\hat{\sigma}_k^{\,2} = \mathbf{v}_k^{\mathrm{T}} \mathbf{P}_k \mathbf{v}_k \,/\, r_k, \quad r_k = \mathrm{tr}(\mathbf{P}_k \mathbf{Q}_{vv,k}), \; \Sigma_k r_k = n - u \tag{3.2}$$

r_k is the partial redundancy of group k: the share of the degrees of freedom that group contributes. Iterate until every group's a-posteriori factor is one; iterated Helmert coincides with the iterated almost-unbiased estimator and, under normality, with REML. This is lsqvce.engine.vce (WP-12), with the per-observation redundancy numbers (redundancy.m) supplying r_k exactly when the problem is small enough to afford them.

3.2 MINQUE and LS-VCE

Rao's MINQUE (1971) is non-iterative and distribution-free but may return negative components and depends on the a-priori values. LS-VCE (Teunissen and Amiri-Simkooei 2008) casts the estimation of the components as a least-squares problem of its own: it delivers the precision of the components, admits covariance components and a-priori information, and coincides with BIQUE under normality. lsqvce does not implement LS-VCE; where the components' own uncertainty is needed it uses the Fisher information of the Helmert iteration (WP-23). This is a documented limitation (section 9), not an oversight.

3.3 Monte-Carlo trace estimation

At GRACE scale r_k in (3.2) needs the inverse of the full normal matrix -- infeasible. Koch and Kusche (2002) and Kusche (2003) replace the exact trace by Hutchinson's stochastic estimator: for probe vectors $\mathbf{z}$ with independent zero-mean unit-variance entries (Rademacher ±1),

$$\mathrm{E}\{\mathbf{z}^{\mathrm{T}}\mathbf{M}\mathbf{z}\} = \mathrm{tr}(\mathbf{M}), \quad \mathrm{Var} \propto 1/m \text{ for } m \text{ probes} \tag{3.3}$$

Each probe costs one extra solve of the normal system; a few dozen probes give traces good to a percent. lsqvce implements this as vce 'hutchinson' / vceHutchinson with ONE probe generator and an honest RNG contract (WP-36: the seed is recorded in Results.meta), and estimates the REDUNDANCY, not the trace of the wrong matrix (WP-32 fixed exactly that). The probe count and the estimator's spread are user-visible (Options.Probes, Options.Seed).

3.4 Degeneracy and non-negativity

A component driven towards zero means the group carries no independent information ($r_k \to 0$); a negative estimate means sampling noise or a mis-specified model. Options are to fix the component, merge groups, or impose non-negativity (Amiri-Simkooei 2016). lsqvce refuses to estimate a group below a minimum redundancy (Options.MinRedundancy, default 50) and reports the per-pass history rather than clamping silently; the negative-factor clamp and the PSD floor of WP-35/40 are declared policies with warning identifiers, never quiet corrections.

4 Stochastic modelling of correlated satellite data

GRACE-FO observables -- K-band range-rate, kinematic positions -- are strongly correlated in time. lsqvce's model, per observation group, is separable (adjustVCE help "THE MODEL"):

$$\mathbf{C}_g = \sigma_g^2 \, \mathbf{D}_g \, (\mathbf{R}_{t,g} \otimes \mathbf{I}_d) \, \mathbf{D}_g \tag{4.1}$$

$\mathbf{D}_g$ is a FIXED per-epoch scaling supplied by the loader (the kinematic-orbit epoch covariance, say); $\mathbf{R}_{t,g}$ the temporal correlation, the same on each of the d axes, ESTIMATED; σ_g^2 the variance component of section 3. No cross-axis correlation is what makes the whitener separable and cheap (separableWhiten).

4.1 Whitening by filters, and the one thing to get right

The temporal law is applied as a whitening filter -- an AR/ARMA filter fitted to the residual autocovariance (residualAcv, selectArAcv, levinsonDurbin), or the exact Toeplitz factor built from an estimated PSD (psd2acv, toeplitzCofactor). A filter restarts its state at every batch, so the first p rows of a batch are transient. Overlapping batches by the filter memory p and discarding the overlap makes the batched result identical to filtering the whole arc; measured (BlockAccumulator help): 1.6e-15 relative error with the overlap, 1.3e-3 without -- a factor 10^{12} that fails silently. The overlap is therefore automatic (BatchOverlap = p) and not a knob.

4.2 The two loops

The SEPARATE loop (Options.Model = "vce") alternates: solve under the current covariance; re-estimate $\mathbf{R}_{t,g}$ from whitened residuals; estimate σ_g^2 by (3.2); push both back into the loaders as a covariance override; repeat until every stream is white (Ljung-Box, Options.WhiteAlpha) and every factor is one (Options.VceTol, OuterTol). The POOLED loop ("pooled") estimates one spectrum over all streams per pass. The JOINT loop ("ellmer") is Ellmer's (2018, §6.5) ten-step co-estimation of PSD amplitudes and keyed per-arc variance factors with Hutchinson probes -- the ITSG-Grace2018 scheme (Kvas et al. 2019) -- as adjustVCEjoint (WP-13/33/34, through fit since WP-56c). The two can converge to DIFFERENT models because amplitudes and scales multiply the same covariance; the toolbox documents this coupling and the production guidance of WP-40 (open loop or at most three damped passes) instead of pretending one answer.

4.3 Non-stationary and background-model noise

Ellmer (2018) propagates the star-camera orientation covariance into the K-band antenna-offset correction, giving a non-stationary noise model recomputed every month and separating stationary ranging noise from the non-stationary

correction -- 27-46 % lower noise than the official centres' solutions at 300 km smoothing (Kvas et al. 2019). Kvas and Mayer-Gürr (2019) derive stationary covariance models for the atmosphere-ocean background-model error and weight them by VCE. In lsqvce the per-epoch scaling $\mathbf{D}_g$ is the hook for the first (epochCovariance in the data contract), a covariance group with its own component the hook for the second.

5 Outliers: screening and robust estimation

Baarda's data snooping tests the normalised residual $w_i = v_i / (\sigma_0 \sqrt{q_{vv,ii}})$ against N(0,1) and removes or downweights the largest exceedance; its reliability theory gives the minimal detectable bias (MDB), and Förstner (1983) extends it to the Gauss-Markov model. lsqvce's diagnose computes w, the redundancy numbers r_i, MDB and reliability per observation; snoop screens block-natively (WP-15); report turns them into a five-level verdict. Two measured traps are documented in README: Pope's τ goes blind under contamination, and the cure has its own cliff -- which is why the verdict never rests on $|w|$ or r alone.

M-estimation replaces rejection by iteratively reweighted least squares with an influence function -- Huber (quadratic core, linear tails), IGG3 (keep / downweight / reject zones), bisquare (redescending). This is lsqvce.engine.robust (WP-25), reached through fit with Options.Robust = "irls" on the in-memory kinds (WP-56b). Robustness and VCE interact: outliers inflate the components; screen first, then estimate, or use the robust arc-factor mean of the joint loop (Options.MeanType).

6 Spherical-harmonic gravity recovery

The disturbing potential is expanded in fully normalised spherical harmonics up to degree and order L_{max}; the common unknowns are the Stokes coefficients. The design matrix holds the partial derivatives of the observable -- range-rate, positions, potential differences -- with respect to each coefficient, i.e. a spherical-harmonic synthesis of the functional. Production processing (ITSG: Kvas et al. 2019) uses variational equations over daily arcs with arc-local nuisance parameters -- exactly the block-arrow structure of section 1.2 -- and stacks the arcs.

lsqvce carries a physically structured closed-loop oracle for this: shDesign (WP-48) generates deterministic SH partials with realistic column correlation, and demo_gravity (WP-51/53) recovers planted coefficients from a GRACE-like, ragged, three-stream fixture with measured numbers pinned (σ_0 0.9998, stream ratios 0.996 / 1.000 / 1.019, planted anomaly recovered at 1.33 σ). That oracle, not a Gaussian design, is the reference authority for the estimator (HANDOVER §6.1.4).

7 Error handling

Two kinds of error meet in an adjustment toolbox: numerical -- floating-point arithmetic destroying information -- and software -- a caller violating a contract. lsqvce treats both under one principle:

> Guards are not fixes. A guard that turns a wrong number into a plausible one has hidden a defect; a guard that turns it into a refusal has reported one. lsqvce refuses.

7.1 Catastrophic cancellation in $\mathbf{v}^T\mathbf{P}\mathbf{v}$

The quadratic form can be obtained without re-forming residuals from quantities the accumulation already holds:

$$\mathbf{v}^T\mathbf{P}\mathbf{v} = \mathbf{l}^T\mathbf{P}\mathbf{l} - \hat{x}^T\mathbf{A}^T\mathbf{P}\mathbf{l} \tag{7.1}$$

When the fit is good the two terms are large and nearly equal and their difference loses most of its digits -- it can even come out slightly negative (Higham 2002, ch. 1). Three responses exist. Clamping to zero fabricates a plausible variance; iterative refinement of the solution masks the conditioning that caused the loss; detecting the loss and returning NaN reports it. lsqvce does the third on BOTH backends (WP-49 normal path: the authoritative $\mathbf{v}^T\mathbf{P}\mathbf{v}$ comes from the residual pass, and cancellation yields NaN under ONE identifier per namespace; WP-50 QR path: the merge-fold cancellation that produced $\sigma_0 = 0$ was found and is guarded by a measured three-anchor threshold with a rotation backstop). A NaN σ_0 is a result: it says the problem's conditioning exceeded what double precision can carry through this formula.

7.2 Factorisation at the singular boundary

Whether dpotrf reports failure on a to-eps-singular matrix depends on the BLAS/LAPACK build: the trailing pivot rounds to ±tiny. A test or a code path that asserts a factoriser's verdict AT the boundary is therefore asserting the library, not the mathematics. Pinned after a live catch (R4 §11.8, 2026-08-30): assert the eigenvalue or rank statement, keep factoriser assertions to clearly definite regimes, and treat regime-dependent behaviour as such (the WP-47 REGIME doctrine).

7.3 Contracts fail fast and name the rule

Every public function validates its inputs at the signature with typed arguments blocks (WP-39); cross-argument rules are checked in the body under lsqvce:<unit>:<rule> identifiers, one identifier per rule (the b1-r3 rule). fit validates the whole data declaration BEFORE any day is read, so a typo costs a second, not a pass over a month (DATA_CONTRACT §3.5). Unknown struct fields are errors on purpose. Errors raised while reading a day carry the loader's identifiers with the user's own error attached as the cause.

7.4 Warnings are inventory, not noise

The CI gate refuses a run whose warning inventory gained an identifier nobody declared (WP-47 rule 7). A test that provokes a warning on purpose mutes exactly that identifier for its own duration. The consequence for the toolbox: every warning identifier is a documented event with a documented cause, or it does not exist.

7.5 Determinism and provenance

The residual pass re-reads the data; a loader that is not deterministic puts its non-determinism into σ_0. Every Results carries an immutable meta record -- toolbox version, MATLAB release, UTC timestamp, every RNG seed resolved, the resolved option set (WP-44) -- so a file can answer "which run produced this" a year later.

8 Implementation: the workflow in lsqvce

The user surface is four names (R4, decisions R4-D1/D4): a data struct describing the problem (DATA_CONTRACT.md), lsqvce.Options for what to do, lsqvce.fit as the one verb, lsqvce.Results with report(). Below it the engines and kernels of the tier census (docs_src/tiers.csv) do the work:

1. **Declare.** data.common states the common unknowns ONCE; data.systems(k) points to buildDay -- the one function the user writes, returning A, B, l (O − C), t for a day, raw and unwhitened.
2. **Build loaders.** fit wraps each buildDay through makeLoader: the meta probe (which arcs a day carries, per day -- ragged schedules are native, WP-52), arc slicing, a single-day cache, the whitener from the covariance spec.
3. **Accumulate.** BlockAccumulator visits system, day, observation type, arc; whitens each arc with the batch overlap of §4.1; accumulates into the (nc+nl) system; after a system's days pre-eliminates its local block by (1.5). Parallel axes that pay: days within a system (DistributeDays, WP-62), arcs in the redundancy pass, systems. Never within an arc (filter state), never inside the dense solve (BLAS).
4. **Solve.** Cholesky or QR on the reduced common system; datum defect detection; declared regularisation.
5. **Residual pass.** The data is read a second time (determinism!); residuals, whitened residuals, the authoritative $v^T Pv$ (§7.1), redundancy numbers exact or by Hutchinson probes.
6. **Iterate.** Model "vce" / "pooled" / "ellmer" re-estimate the stochastic model and the components and push them back into the loaders as a covariance override (opt.covOverride.bySys, resolved by the ONE merge applyCovOverride -- G-06), until white and balanced.
7. **Judge.** diagnose and report: per-observation w, r, MDB; the five-level verdict; coefficients() with confidence bounds; everything reachable on the Results for experts (R.acc, R.loaders, R.noise).

Cost model. Every pass reads each day twice; the VCE loop multiplies by the pass count. At global SH scale the redundancy / VCE data pass is where the hours go (about 4 h per iteration at degree 90 serial; 30-40 min with 16 workers, BlockAccumulator help) -- build the workflow around parallelising THAT axis.

Kind of data	Fields present	Engine	Options.Model
Streaming (GRACE-FO)	systems, common, days	adjustBlock / adjustVCE / adjustVCEjoint	gls, vce, pooled, ellmer
In-memory matrices	A, l, (p)	adjust / robust	gls (+Robust)

Kind of data	Fields present	Engine	Options.Model
Batched callback	batchFcn, numUnknowns, numBatches, (x0)	adjust / robust / solveNonlinear	gls (+Robust)

9 Limitations of the method and of the implementation

- Helmert / redundancy-contribution VCE estimates variance components of UNCORRELATED groups. Covariance components between groups (LS-VCE) are not implemented.
- The separable model (4.1) assumes one temporal law shared by the axes of a stream and no cross-axis correlation; a vector-AR model (varWhiten, whittle) exists for the multi-axis case but is not the default.
- Monte-Carlo redundancy is unbiased but noisy: its spread scales as $1/\sqrt{m}$ in the probe count. Components stable to a percent need tens of probes; the seed is recorded so a run can be reproduced exactly.
- The joint (Ellmer) and the separate loop can converge to different stochastic models; the alternation can be bistable (F-15). Production guidance -- open loop or at most three damped passes -- is documented; the monotone / trust-region cure is a backlog item (WP-42).
- Cancellation in $v^{T}Pv$ is REPORTED (NaN), not repaired; a problem whose conditioning exceeds double precision through this formula needs rescaling or the QR path, which the diagnostics recommend but do not switch to silently.
- A NaN σ_0, a group below the redundancy floor, a non-converged loop are results with warning identifiers -- the toolbox never returns a plausible substitute in their place.

10 References

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